

Network Troubleshooting Guide

1. Network Connectivity Basics

Stable network connection essential for smooth application performance. Check bandwidth, latency, and packet loss. Problems range from poor WiFi signal to ISP issues. Wired connection (Ethernet) more stable than WiFi. Mobile hotspot acceptable but less reliable than fixed broadband.

Minimum Network Requirements:

- Download Speed: Minimum 5 Mbps (10 Mbps recommended)
- Upload Speed: Minimum 2 Mbps (5 Mbps recommended)
- Latency (Ping): Under 100ms (ideal <50ms)
- Packet Loss: Less than 1%
- DNS Resolution: Under 50ms
- Jitter: Less than 20ms variation

Testing Your Connection:

Use speedtest.net to measure connection. Run test from wired connection for baseline. Test multiple times throughout day. Morning speeds differ from evening (peak usage). Video calls require minimum 4 Mbps per participant. File uploads need good upload speed.

2. WiFi & Router Issues

WiFi problems most common cause of connectivity issues. Signal strength, interference, and distance from router all matter. Walls and obstacles weaken signal. Microwaves and cordless phones cause interference. Changing channels can improve connection.

WiFi Troubleshooting Steps:

1. Move closer to router (maximum 30 feet for optimal signal)
2. Restart router (power off 30 seconds, power on)
3. Check WiFi signal strength (target -30 to -67 dBm)
4. Switch WiFi channel (channels 1, 6, 11 least interference on 2.4GHz)
5. Use 5GHz band if available (faster, less interference)
6. Update router firmware from manufacturer website
7. Check router logs for error messages

Router Configuration:

Access router admin panel (usually 192.168.1.1). Set strong WiFi password using WPA3 encryption (WPA2 if WPA3 unavailable). Disable WEP (obsolete, insecure). Enable automatic channel selection. Reduce transmit power if interference issues persist. Separate SSIDs for 2.4GHz and 5GHz bands can help optimize connections.

3. ISP Issues & DNS Problems

Internet Service Provider (ISP) issues cause widespread connectivity outages. Check ISP status page or call ISP support. DNS (Domain Name System) translates website names to IP addresses. DNS issues cause 'site cannot be reached' errors.

DNS Troubleshooting:

Try alternative DNS: Use 8.8.8.8 and 8.8.8.1 (Google DNS) or 1.1.1.1 and 1.0.0.1 (Cloudflare). Windows: Settings → Network → Change Adapter Options → Properties. Linux: Edit /etc/resolv.conf. Mac: System Preferences → Network → Advanced → DNS.

Network Diagnostics Commands:

Windows: ipconfig (view IP config), ping 8.8.8.8 (test connection), tracert google.com (trace route)

Mac/Linux: ifconfig (view IP config), ping -c 4 8.8.8.8 (test connection), traceroute google.com (trace route)

Persistent Connectivity Issues:

If issues persist after troubleshooting, try wired Ethernet connection to eliminate WiFi. Check Device Manager (Windows) or System Report (Mac) for driver issues. Uninstall and reinstall network drivers. If wired connection works but WiFi doesn't, issue is WiFi hardware or driver. Contact ISP for line tests if wired connection still fails.

Network Support Resources:

Network Issues Team:

Email: network@company.com

Phone: 1-800-NETWORK

Chat: Available 24/7 in support portal

Status Page: status.company.com

Response Time: Within 1 hour for outages